EXP 28

```
clc;
clear;
close all;

% DSSS Communication System
% PN Code Generation, Spreading and Signal Recovery

% Step 1: Generate input data
data = [1 0 1 1 0 1 0 0];
N = length(data);

% Convert binary data to bipolar form
data_bipolar = 2*data - 1;

% Step 2: Generate PN sequence using 3-bit LFSR
reg = [1 1 1];
pn = zeros(1,7);

for k = 1:7
    pn(k) = reg(3);
    feedback = xor(reg(3),reg(1));
    reg = [feedback reg(1:2)];
end

% Convert PN sequence to bipolar form
pn_bipolar = 2*pn - 1;
L = length(pn);

% Step 3: Spread the data
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spread_signal = zeros(1,N*L);

for k = 1:N
   idx = (k-1)*L + (1:L);
   spread_signal(idx) = data_bipolar(k)*pn_bipolar;
end

% Step 4: Add AWGN noise
SNR_dB = 10;
signal_power = mean(spread_signal.^2);
noise_power = signal_power / (10^(SNR_dB/10));
noise = sqrt(noise_power)*randn(size(spread_signal));

received_signal = spread_signal + noise;

% Step 5: Despread and recover data
recovered_data = zeros(1,N);

for k = 1:N
   idx = (k-1)*L + (1:L);
   received_chips = received_signal(idx);

   correlation = sum(received_chips .* pn_bipolar);

   if correlation > 0
      recovered_data(k) = 1;
   else
      recovered_data(k) = 0;
   end
end
```

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% Step 6: Display results
disp('Original Data:');
disp(data);

disp('Generated PN Sequence:');
disp(pn);

disp('Recovered Data:');
disp(recovered_data);

if isequal(data,recovered_data)
   disp('Data Recovery Successful');
else
   disp('Some Bits Were Recovered Incorrectly');
end

% Step 7: Plot signals
figure;

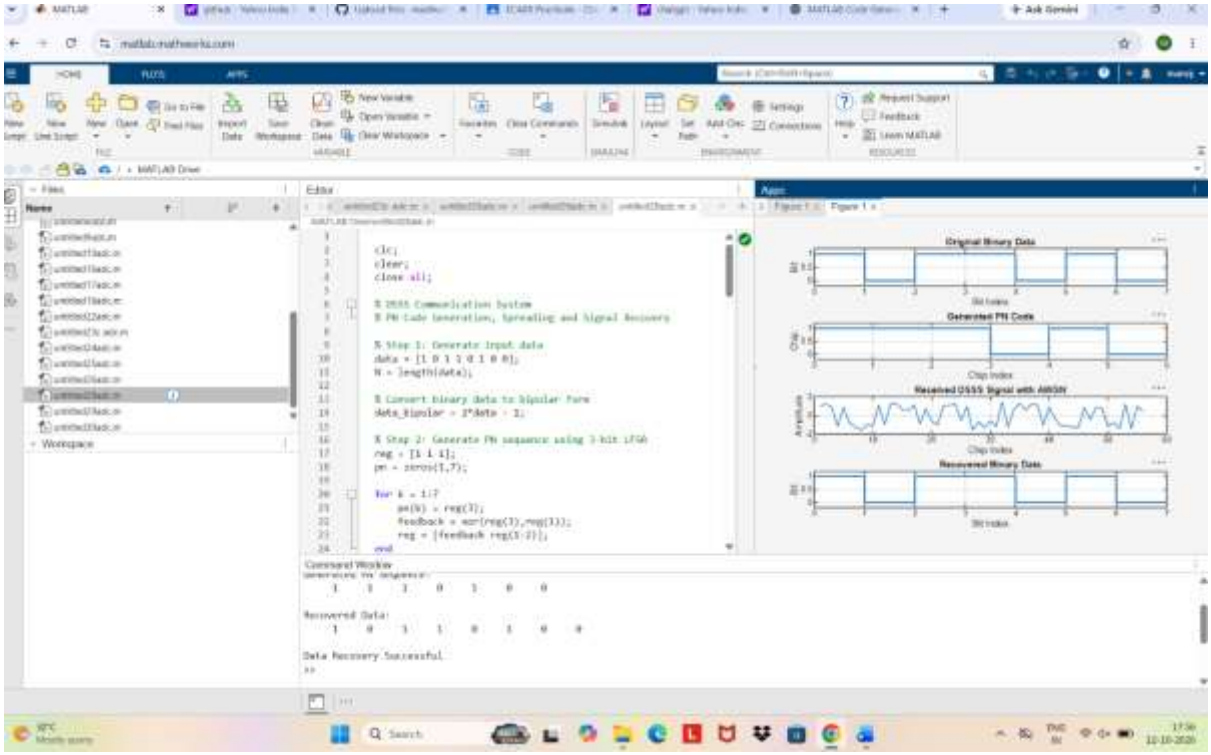
subplot(4,1,1);
stairs(0:N-1,data,'LineWidth',1.5);
title('Original Binary Data');
xlabel('Bit Index');
ylabel('Bit');
ylim([-0.2 1.2]);
grid on;

subplot(4,1,2);
stairs(0:L-1,pn,'LineWidth',1.5);
title('Generated PN Code');
xlabel('Chip Index');
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ylabel('Chip');
ylim([-0.2 1.2]);
grid on;

subplot(4,1,3);
plot(received_signal,'LineWidth',1);
title('Received DSSS Signal with AWGN');
xlabel('Chip Index');
ylabel('Amplitude');
grid on;

subplot(4,1,4);
stairs(0:N-1,recovered_data,'LineWidth',1.5);
title('Recovered Binary Data');
xlabel('Bit Index');
ylabel('Bit');
ylim([-0.2 1.2]);
grid on;
```


Original Binary Data
Bit
Bit Index
Generated PN Code
Chip
Chip Index
Received DSSS Signal with AWGN
Amplitude
Chip Index
Recovered Binary Data
Bit
Bit Index
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clear;
close all;
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% PN Code Generation, Spreading and Signal Recovery
% Step 1: Generate input data
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for k = 1:7
pn(k) = reg(3);
feedback = xor(reg(3),reg(1));
reg = [feedback reg(1:2)];
end
1 1 1 0 1 0 0
Recovered Data:
1 0 1 1 0 1 0 0
Data Recovery Successful
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